

Correcting for finite statistics effects in a quantum steering experiment

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(Received 10 November 2023; accepted 22 January 2025; published 16 May 2025)

Verifying entanglement between parties is essential for creating secure quantum communication. However, finite statistics can lead to false positive outcomes in any tests for entanglement. Here, we introduce a one-sided device-independent protocol that corrects for apparent signaling effects in experimental probability distributions, caused by statistical fluctuations and experimental imperfections. We use semidefinite programming to identify the optimal inequality, for our experimental probability distribution, without resource-intensive tomography. Our protocol is numerically and experimentally analyzed in the context of random, misaligned measurements, correcting apparent signaling where necessary. Our results show a significantly higher probability of violation than existing state-of-the-art inequalities. This study demonstrates the power of semidefinite programming for entanglement verification and brings quantum networks closer to practical applications.

DOI: [10.1103/PhysRevResearch.7.023156](https://doi.org/10.1103/PhysRevResearch.7.023156)

Introduction. Quantum networks have applications across many sectors, from secure communications [1,2] to distributed sensing [3] and the development of a quantum internet [4,5]. Thanks to recent technological advances, such as ultra-bright photon sources and sophisticated detectors, quantum information processing tasks in quantum networks are now becoming a reality [4,6,7]. This has been successfully demonstrated in free-space and fiber-based quantum networks [8,9]. The most rigorous way to verify nonlocality in networks are Bell tests, which are fully device independent and treat measurement devices as black boxes [10–13]. However, despite today's technological advancements, Bell tests remain demanding for realistic quantum networks.

Thus, quantum steering—a relaxation of Bell tests—has received a substantial amount of attention [14–16], thanks to its noise tolerance and loss robustness [17,18]. Steering

based tasks are one-sided device independent (1s-DI), meaning that one party's measurement devices are trusted, while the other party's devices are untrusted and treated as a black box. This allows the trusted party, Bob, to perform quantum state tomography and gain full information about his state, while the untrusted party, Alice, is limited to her conditional measurement outcomes.

Experimental demonstrations of such nonlocality tests assume that the collected statistics are not biased to any experimental imperfections and represent a fair sample of the true underlying distribution of the shared quantum states. Additionally, reduced transmission rates in long-distance quantum communication, in combination with imperfect detector technology, can result in scenarios where only limited data can be collected, increasing the effect of finite statistics. For practical applications of nonlocality in quantum networks, these limitations need to be reduced. This either requires careful experimental designs [19,20] and the use of advanced detection technology, or demands more sophisticated numerical approaches.

Here, we develop a resource-efficient protocol that allows us to demonstrate steering by numerically correcting for

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finite statistics effects. Our protocol is based on the steering robustness (SR) [21], a quantifier of quantum steering that uses semidefinite programming (SDP) and resource-intensive full-state tomography on Bob's side. Using this measure rather than the binary test of steering inequalities informs us about the noise robustness of an entangled state [22]. In comparison to the SR, our protocol—the adapted steering robustness (ASR)—is resource minimal, as it does not require tomography on either party. Furthermore, it numerically searches for the tightest, optimal, steering inequality for the recorded data and is efficiently solvable on a classical computer. We find, in simulations, that for a maximally entangled state, the ASR outperforms other compared steering inequalities when Bob and Alice perform two misaligned orthogonal measurements each. For the three settings per party case, the ASR is optimal.

We experimentally test our protocol on a robust photonic quantum processor [23] with the stability to perform 10 000 random orthogonal measurements per side. Our method requires the shared quantum correlations to be nonsignaling [24–26], which is a built-in constraint for any steering criterion. Without satisfying this constraint, there is the potential for false positives to occur and this must be carefully considered. Owing to finite statistics, experimental demonstrations rarely fully satisfy this principle [27,28] and our analysis involves a numerical optimization method. Our 1s-DI nonsignaling model algorithm (NSMA) finds the closest, according to a chosen order of merit, nonsignaling statistics, that we assume to be the underlying behavior of our experiment. After applying the NSMA to our experimental statistics, we compare the ASR to other well-established quantum steering inequalities [29–32] and find that the ASR performs significantly better than all other inequalities.

Results. In a steering scenario, one assumes that two parties, Alice and Bob, share a quantum state ρ_{AB} . Bob tries to establish if the state is entangled, without trusting Alice's measurement outcomes. In each iteration of the protocol, Bob receives his part of the shared state and announces randomly chosen measurement settings $x \in \{1, \dots, n\}$ for Alice to perform locally on her part of the state. Then, Alice declares her corresponding measurement outcome a to Bob. Finally, Bob has access to his collection of postmeasurement states $\{\varrho_{a|x}\}_{a,x}$ along with Alice's conditional probabilities $\{p(a|x)\}_{a,x}$. This information is expressed in the set of states, known as the state assemblage $\sigma_{a|x} = p(a|x)\varrho_{a|x}$. Alice's measurements are described by a positive-operator valued measure (POVM) $\{A_{a|x}\}_a$ for each x , i.e., $\sum_a A_{a|x} = \mathbb{1}$ and $A_{a|x} \geq 0$ for each a , where $\mathbb{1}$ is the identity, and ρ_{AB} is the state shared between Alice and Bob. Given this formalism, members of the assemblage set can be described by $\sigma_{a|x} := \text{tr}_A[(A_{a|x} \otimes \mathbb{1})\rho_{AB}]$. Bob can use the assemblage to test if Alice's results can be explained by a local hidden state (LHS) model [14]. If such an LHS model does not exist, the shared state ρ_{AB} is entangled. Thus, Alice can steer his system via her measurements. In this scenario, Bob is required to have complete knowledge of his postmeasurement state $\varrho_{a|x}$. Besides verifying if a shared state is steerable, one can determine how much noise, ubiquitous in practical demonstrations, can be added before a specific assemblage becomes unsteerable. This is quantified using the SR [21], which is a measure of how much noise can be

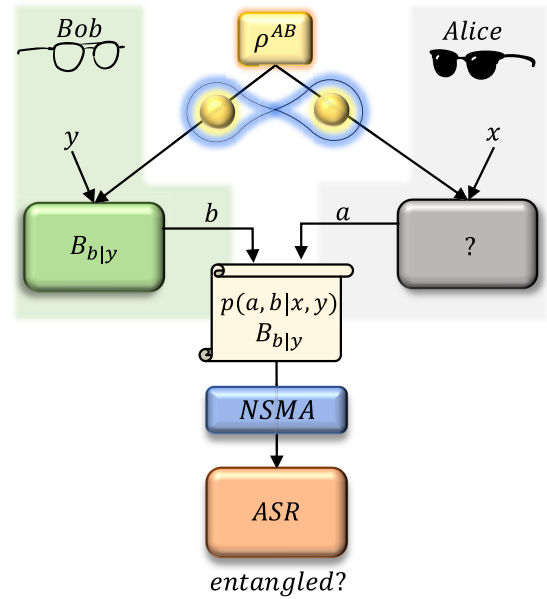


FIG. 1. Adapted steering robustness protocol. A trusted party (Bob) wants to verify entanglement in a shared state ρ^{AB} with an untrusted party (Alice) using measurement settings y and x , respectively. Bob performs a POVM $B_{b|y}$, while Alice announces her measurement outcomes a . For statistical reasons, they repeat the experiment to form a joint probability distribution $\{p(a, b|x, y)\}_{a,b,x,y}$. To correct for the effects of finite statistics, the NSMA is implemented, finding the closest nonsignaling probability distribution that is realizable with Bob's measurements. The ASR is then calculated via a semidefinite program and used to verify entanglement.

added to a state before it becomes unsteerable [16]. Thus, any state with SR greater than zero is steerable. To calculate SR, semidefinite programming (SDP) [22], a convex optimization algorithm, over $\sigma_{a|x}$ is used to find the optimal witness,

$$\begin{aligned} \text{SR} &= \max_{\{F_{a|x}\}} \sum_{a,x} \text{tr}[F_{a|x}\sigma_{a|x}] - 1 \\ \text{s.t. } &\mathbb{1} - \sum_{a,x} D_\lambda(a|x)F_{a|x} \geq 0 \quad \forall \lambda \\ &F_{a|x} \geq 0 \quad \forall a, x, \end{aligned} \quad (1)$$

where $\{F_{a|x}\}_{a,x}$ is a set of Hermitian operators and $D_\lambda(\cdot|x)$ is a deterministic probability distribution.

This method gives an optimal steering inequality, but it requires quantum state tomography on Bob's part of the state, demanding a precise calibration of measurement devices. Alternatively, we consider a less resource-intensive case, where Bob is limited to a fixed set of measurements, described by a POVM $\{B_{b|y}\}_b$ for each y , i.e., $\sum_b B_{b|y} = \mathbb{1}$ and $B_{b|y} \geq 0$ for each b , and collects his conditional probabilities to construct the joint probability distribution $\{p(a, b|x, y)\}_{a,b,x,y}$. Therefore, we reformulate the steering robustness protocol using the joint probability distribution, instead of the state assemblage. A schematic figure of the protocol is presented in Fig. 1.

Adapted steering robustness. In our scenario, we have access to the joint probability distribution $\{p(a, b|x, y)\}_{a,b,x,y}$, rather than $\sigma_{a|x}$, and define an adapted steering robustness

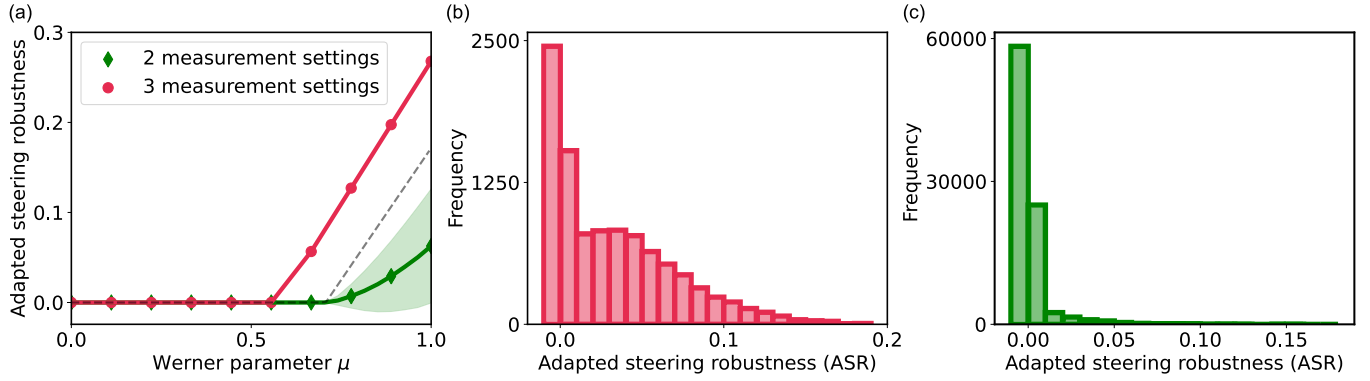


FIG. 2. (a) Simulated mean adapted steering robustness $\overline{ASR}$ as a function of mixing parameter μ for two (diamonds, green) and three (circles, red) random measurement settings per site. For three settings, Alice selects three orthogonal settings from a QU-random distribution, and Bob's measurements are fixed to $\sigma_x, \sigma_y, \sigma_z$. For two settings, Alice selects pairs of orthogonal settings from the same QU distribution, while Bob randomly selects pairs of Pauli measurements. Both cases are repeated for $n = 50$ samples per data point. The data points represent the mean ASR values and one standard deviation (shaded area). For comparison, the dashed line shows the steering robustness SR for two measurement settings on Alice's side, chosen from the same QU distribution, and tomography using the Pauli measurements on Bob's side. Owing to full state reconstruction by Bob, the SR achieves the maximum ASR value. (b) and (c) show histograms of adapted steering robustness (ASR) samples, computed using an experimental probability distribution after applying the NSMA algorithm. (b) Three settings case, 10 000 random mutually unbiased measurements, 75.5% of the ASR values are nonzero, indicating steerability. (c) Two settings case, 90 000 pairs of random measurements, and 32.6% of ASR are nonzero.

(ASR),

$$\begin{aligned} ASR = \max_{\{\alpha_{xy}^{ab}\}} \sum_{a,x} \alpha_{xy}^{ab} p(a, b|x, y) - 1 \\ \text{s.t. } 1 - \sum_{a,x} D_\lambda(a|x) \sum_{b,y} \alpha_{xy}^{ab} B_{b|y} \geq 0 \quad \forall \lambda \quad (2) \\ \sum_{b,y} \alpha_{xy}^{ab} B_{b|y} \geq 0 \quad \forall a, x, \end{aligned}$$

where $\{\alpha_{xy}^{ab}\}_{a,b,x,y}$ are real coefficients and $B_{b|y}$ are Bob's trusted measurements. We note that there are no limits on the values $\{\alpha_{xy}^{ab}\}_{a,b,x,y}$ can take for a nonsignaling scenario (see part A of the Supplemental Material [33]). The steering inequality, based on Eq. (2) is

$$\sum_{a,x} \alpha_{xy}^{ab} p(a, b|x, y) - 1 \leq 0, \quad (3)$$

where $\{p(a, b|x, y)\}_{a,b,x,y}$ are joint probabilities and $\{\alpha_{xy}^{ab}\}_{a,b,x,y}$ are real coefficients calculated via the SDP defined in Eq. (2).

Here, we consider a two-qubit state, i.e., a Werner state $\rho_\mu = \mu |\psi_s\rangle \langle \psi_s| + \frac{(1-\mu)}{4} \mathbb{1}_4$, which is a probabilistic mixture of a maximally entangled singlet state $|\psi_s\rangle = \frac{1}{\sqrt{2}}(|10\rangle - |01\rangle)$ with symmetric noise parameterized by the mixing probability $\mu \in [0, 1]$ [34]. We choose Alice and Bob's measurement settings to be $A_i = \vec{a}_i \cdot \vec{\sigma}$ and $B_i = \vec{b}_i \cdot \vec{\sigma}$, respectively, where $\vec{\sigma} = (\sigma_x, \sigma_y, \sigma_z)$ is a vector composed of the Pauli matrices, $\vec{a}_i \in \mathbb{R}^3$ and $\vec{b}_i \in \mathbb{R}^3$. We chose Bob's measurement settings along n random mutually unbiased bases (MUBs), quasi-uniformly (QU) distributed along the Bloch sphere (see part B of the Supplemental Material [33]).

Firstly, we numerically simulate the ASR as a function of the mixing parameter μ [Fig. 2(a)]. Our results demonstrate steering on average with positive $\overline{ASR}$ values for $\mu \geq \frac{1}{\sqrt{2}}$ and

$\mu \geq \frac{1}{\sqrt{3}}$ for two and three measurement settings per side, respectively. These values match standard results for the SR when considering Werner states [22].

For maximally entangled states with $\mu = 1$ and three measurement settings per side, we obtain a maximum ASR_3 value of 0.2679, equivalent to the SR value. By reducing the measurements to two settings per side—the minimal set size—we obtain $ASR_2 = 0.0627$ for $\mu = 1$, when averaged over all combinations of measurements. For two perfectly aligned measurements per side, Eq. (3) can take a maximum value of 0.1716. We note that in this scenario, the ASR_2 value is identical to Eq. (1). We also note that $SR_2 \geq \overline{ASR}_2$ though $\max(ASR_2) = SR_2$ as the SR has access to full tomographic information on Bob's side, so when measurements are random, SR will perform better, on average, than the ASR.

We implemented the steering protocols using a state-of-the-art silicon photonic chip. The entangled photon pairs at 1550 nm are generated via spontaneous four-wave mixing in the two spiral-waveguide sources that are pumped by a 1550 nm fiber-coupled continuous-wave laser with an output power (after fiber) of 300 mW. The generated state is verified using quantum state tomography at several stages throughout the experiment—in each case, we achieved a fidelity of ca. 93% with the singlet state $|\psi_s\rangle$. Alice and Bob's projective measurements are implemented by thermo-optic phase shifters on chip together with coincidence detections using fiber-coupled superconducting nanowire single-photon detectors off-chip. See Fig. S1 within the part B of the Supplemental Material [33] for a schematic of the chip. The coincidence data was compiled into a joint probability distribution $\{p_{\text{exp}}(a, b|x, y)\}_{a,b,x,y}$ together with Bob's measurements $\{B_{b|y}\}_{b,y}$ to calculate the ASR [Eq. (2)]. For this, we used an SDP encoded using CVX [35], in addition to open-source codes from Refs. [36,37]. The experimental data is available here [38].

The ASR, like any steering criterion, is ill-defined for signaling distributions. Applying the ASR algorithm to our experimental probability distribution leads to unbounded values for all random measurement settings in both the two and three measurements per side cases. It is noteworthy that many steering criteria can be evaluated even for signaling data without noticing such behavior, although the fact that one uses signaling data would contradict the constraints that go into deriving the criterion in the first place. Here, the choice of the interval of the parameters $\{\alpha_{xy}^{ab}\}$ is such that it clearly highlights the problem that arises if one does not handle signaling data appropriately. Thus, we propose a method for finding a nonsignaling approximation of the observed data, that is tailored for the steering setup.

Correcting for signaling. Quantum nonlocal correlations have to comply with the NS principle. Thus, the probability distributions of outputs of any subset of parties are independent of the remaining parties' inputs. Because of the finite nature of any experimental statistics, $\{p_{\text{exp}}(a, b|x, y)\}_{a,b,x,y}$ can never strictly satisfy NS conditions [27]. A probability distribution that exhibits signaling does not generate a valid assemblage and, hence, one should refrain from evaluating steering criteria on it. We emphasize that not satisfying the NS principle is different from closing well-known loopholes, such as the locality loophole [39] and the detection loophole [40]. However, we highlight that even in a loophole-free experiment, the effects of finite statistics remain and must be corrected.

Previous methods of finding the closest NS probability distribution minimized the relative entropy between a given signaling distribution and an NS distribution [41]. The NS distribution found is physical but it may not be realizable with Bob's measurements $B_{b|y}$. Thus, we propose a one-sided device independent non-signaling model algorithm (NSMA) that finds the closest NS probability distribution by minimizing the amount of signaling $t \geq 0$ one needs to add to a signaling distribution, given $B_{b|y}$, to generate an NS distribution. The algorithm is defined by an SDP,

$$\begin{aligned}
 t = \min_{\{\tilde{\sigma}_{a|x}\}} & \frac{1}{\#x} \sum_{a,x} \text{tr}[\tilde{\sigma}_{a|x}] - 1 \\
 \text{s.t. } & \text{tr}[\tilde{\sigma}_{a|x} B_{b|y}] - p_{\text{exp}}(a, b|x, y) \geq 0 \quad \forall a, b, x, y \\
 & \tilde{\sigma}_{a|x} \geq 0 \quad \forall a, x \\
 & \sum_a \tilde{\sigma}_{a|x} = \sum_a \tilde{\sigma}_{a|x'} \quad \forall x \neq x',
 \end{aligned} \quad (4)$$

where $\#x$ is the number of Alice's measurement settings and $\tilde{\sigma}_{a|x} = (1+t)\sigma_{a|x}$, where $\sigma_{a|x}$ is the NS assemblage we are searching for. The corresponding NS probability distribution is $p_{\text{exp}}^{\text{NS}}(a, b|x, y) = \frac{\text{tr}[\tilde{\sigma}_{a|x} B_{b|y}]}{1+t}$, see part C within the Supplemental Material [33] for further details on the derivation of the NSMA. We then ensure the NSMA is not biased towards indicating steerability—in that its effects do not increase the probability of an $\text{ASR} > 0$. We used Monte Carlo methods to simulate the impact of finite statistics followed by the NSMA on the ASR for sets of states and measurements on the boundary conditions of the LHS model. We found that the

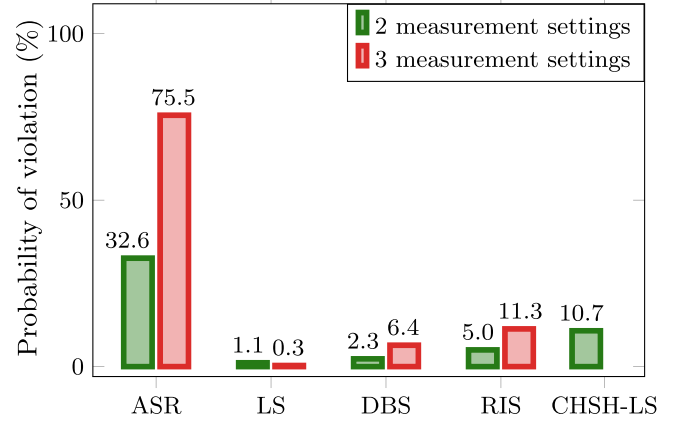


FIG. 3. Experimental probabilities of violation for two (green) and three (red) orthogonal measurements per site for different criteria. Alice's measurement settings are chosen from a quasi-uniform random distribution on the surface of the Bloch sphere, whilst Bob's measurement settings are fixed along $\sigma_x, \sigma_y, \sigma_z$. The probabilities of violation are calculated using the nonsignaling (NS) probability distribution, closest to our experimental probabilities, which is obtained using the NSMA. We compare the ASR probabilities of violation to comparable inequalities: linear steering (LS), dimension-bounded steering (DBS), rotationally-invariant steering (RIS) and Clauser-Horne-Shimony-Holt-like steering (CHSH-LS).

NSMA underestimates the steerability here. See part D within the Supplemental Material [33] for details.

We applied this method to our experimental probability distribution $\{p_{\text{exp}}(a, b|x, y)\}_{a,b,x,y}$ to find the closest NS distribution $\{p_{\text{exp}}^{\text{NS}}(a, b|x, y)\}_{a,b,x,y}$, for each of the 10 000 randomized measurements. Figure 2(b) shows the distribution of ASR values for three measurement settings per side. Our experimental results show a 75.5% chance of violating the ASR inequality. We find $\overline{\text{ASR}}_{\text{exp-3}} = 0.033$ and $0 \leq \text{ASR}_{\text{exp-3}} \leq 0.1885$. For two measurement settings per side, we consider every combination of pairs of the three MUB measurements. This results in 90 000 samples and $\overline{\text{ASR}}_{\text{exp-2}} = 0.0049$, with values $0 \leq \text{ASR}_{\text{exp-2}} \leq 0.1632$ for all samples [Fig. 2(c)]. We violate the ASR-2 inequality with a probability of 32.6%.

Comparison to other steering inequalities. To assess the strength of the ASR, we calculated the probability of violation of the NS distribution for other well-known steering inequalities: linear steering (LS) [29,30], CHSH-like steering (CHSH-LS) [42], dimension-bounded steering (DBS) [31], and rotationally invariant steering (RIS) [32] (see part E within the Supplementary Material [33]). In Fig. 3, we compare the ASR inequality violation probability [Eq. (3)] to comparable steering inequalities. In part F within the Supplemental Material [33], we present a table comparing the probabilities of violation before and after applying the NSMA to highlight the importance of correcting for the effects of finite statistics.

We find the probability of violating the ASR inequality greatly outperforms all other considered inequalities. In the three settings case, for the rotation-invariant inequalities, DBS-3 and RIS-3, the NS experiment only achieves 6.4% and 11.3% probability of violation respectively and the linear inequality LS-3 performed the worst, having a 0.3%

probability of violation. For two measurements, the probability of demonstrating steering using the other inequalities ranges between 1.1% for LS-2 and 10.7% for CHSH-like steering. Numerical simulations comparing the probability of violation for a maximally entangled state ($\mu = 1$) can be found in part F within the Supplementary Material [33].

Discussion. Our approach to quantifying steering represents an important step towards certifying entanglement for future quantum networks. By harnessing the power of SDPs, we have demonstrated the noise-robustness of our ASR witness. Furthermore, we have shown our approach performs particularly well in scenarios involving fully misaligned measurements and finite statistics, where other inequalities underperform.

In previous experiments, the effects of finite statistics have often been excluded or additional assumptions have been made [10–13]. Without satisfying the nonsignaling constraint, any test that excludes local hidden models cannot genuinely prove that the observed entanglement is not a result of signaling. Here, in our proof-of-principle demonstration, we witness entanglement in a steering scenario, with statistics that are by definition of the NSMA algorithm—nonsignaling. Therefore, we are not required to exclude any data or make any additional assumptions as is often the case.

Additionally, our protocols require minimal assumptions about either party's measurements. On one hand, removing the constraint of full tomography on the trusted party allows us to present optimized inequalities based solely on correlations. On the other hand, we retain the loss and noise robustness of steering protocols. Furthermore, our NSMA is an SDP based method for fitting data into a quantum model. Albeit the method here is presented for quantum steering, it is directly adaptable to quantum state estimation and quantum algorithms [43]. We expect that such a convex-geometric linear method can be favorable over distance-based and entropy-based methods in terms of computational efficiency

and will contribute to the emerging field of finite statistics in quantum networks. Overall, the potential for using experimental techniques to explore the effects of finite statistics is an exciting area of research with many possible applications in quantum information processing [44] and other fields [45,46].

Acknowledgments. We thank Tiff Brydges, Benoît Vermersch, Paul Skrzypczyk, Josh Silverstone, and Jeremy Adcock for useful discussions. We thank Graham Marshall, Callum Wilkes, and Laurent Kling for their contributions to the design and characterization of the silicon photonic chip. This project was supported by the Centre for Nanoscience and Quantum Information (NSQI) and by funding from the European Union's Horizon 2020 research and innovation programme under the Marie Skłodowska-Curie Grant Agreement No. 892242, VERIQ-TAS-QuantERA II Programme under Grant Agreement No. 101017733, the Polish National Science Center (Grant No. 2021/03/Y/ST2/00175), and EPSRC Grants (No. EP/T00097X/1) QUANTIC, No. EP/M024385/1, No. EP/SO23607/1, and No. EP/W003252/1, as well as from CAPES/Brazil and the Swiss National Science Foundation (Ambizione PZ00P2-202179). J.W. acknowledges support from the Natural Science Foundation of China (Grants No. 62235001, and No. 61975001), the Innovation Program for Quantum Science and Technology (2021ZD0301500), the National Key R&D Program of China (2019YFA0308702), Beijing Natural Science Foundation (Z190005, Z220008). X.Q. acknowledges support from the National Natural Science Foundation of China (Grant No. 62075243). J.L.O. acknowledges funding from Royal Academy of Engineering Chair in Emerging Technologies. A.C.S.C. was supported by CNPq/Brazil, Grants No. 308730/2023-2 and No. 409673/2022-6. A.C.O.J. was supported by NCN SONATA BIS Project No. 2022/46/E/ST2/00115.

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